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Editors

Physical Review D

Dear Editors:

I submit the manuscript “Normal-Form Resolution of a Free-Boundary MOTS Saddle-Node in Dynamical Five-Dimensional Gravity” for consideration as a Research Article in *Physical Review D*.

The manuscript addresses a boundary-sensitive question in black-hole horizon theory and numerical relativity. Marginally outer trapped surface (MOTS) pair creation is already understood as a saddle-node mechanism for closed surfaces. Here the surface terminates on a physical brane, so the contact condition changes the domain of the stability operator and the corresponding discrete adjoint projection. The ambient five-dimensional geometry is also generated by a nonlinear Einstein–scalar initial-boundary-value evolution rather than prescribed as an exact metric family. Existing closed-MOTS examples and free-boundary stability or rigidity theory therefore do not by themselves establish that the fold hypotheses are realized in this operator problem.

The principal result is a numerical resolution of that critical structure, verified across three spatial grids and an independent half-timestep evolution in the $SO(3)$ -symmetric sector. An independently constructed parent sequence evolves from no accepted brane-capped MOTS to two opposite-stability branches on three successive spacetime grids. Pseudo-arclength continuation connects the branches through a common critical surface. The principal stability eigenvalue is bracketed through zero with an isolated sign-definite mode; endpoint-eliminated discrete adjoint projections give nonzero transversality and quadratic coefficients; and the invariant proper-area separation follows the expected square-root law. An independent G10 half-timestep evolution repeats the parent and continuation and yields an overlapping critical-time bracket with a free exponent near one half.

The contribution is not a claim to a new universal bifurcation mechanism. It is a controlled numerical realization of the normal form in a dynamical free-boundary MOTS problem whose physical Robin condition, non-self-adjoint operator, and evolved higher-dimensional geometry materially alter the calculation. This intersection of general relativity, horizon theory, extra-dimensional gravity, and computational physics falls directly within the scope of *Physical Review D* and should be of interest to readers working on black-hole dynamics and numerical relativity.

The manuscript carefully limits the inference to finite-resolution, quasi-local evidence: it does not claim a continuum theorem, event-horizon reconstruction, nonsymmetric nonprincipal spectrum, or phase diagram. The supporting numerical records, frozen protocols, source manifests, and tests are publicly archived at the exact published version DOI doi:10.5281/zenodo.22087164, within the version series identified by concept DOI doi:10.5281/zenodo.22085054, with version-controlled source at GitHub. The manuscript is submitted as a regular Research Article and is not part of a joint submission.

Thank you for considering this work.

Sincerely,

Ronald Bibb